MATH210B Homework 1

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1 Problem 2

Let F be the field of fractions of R.

For the forward direction, note that if a matrix a has a zero divisor in the ring $M_n(R)$, then it has a zero divisor in $M_n(F)$, which means $det(a) = 0$ by linear algebra.

Similarly, if $det(a) = 0$ in R, it's 0 in F which means there's a zero divisor $b \in M_n(F)$ by linear algebra. Then, by clearing the denominators of b, we get a zero divisor in $M_n(R)$.

2 Problem 9

We'll use the following lemma in the problem:

Lemma 2.1. Let I be an ideal of a non-zero commutative ring R. Then, $(R/I)[t]$ is isomorphic to $(R[t])/\hat{I}$, where $\hat{I}$ in the second ring is the ideal generated by I.

Proof. Consider $\phi : R[t] \to (R/I)[t]$ mapping every coefficient modulo I. ϕ maps 1 to 1. ϕ is clearly a ring homomorphism since it's simply applying the canonical projection map to every component. It's also clearly surjective. Thus, it suffices to show that $\ker(\phi) = \hat{I}$ to conclude the proof.

Let $f \in \hat{I}$. Then, $f = gx$ for some $g \in R[t]$ and $x \in I$. Then, $x \mid C(f)$, so $f \in ker(\phi)$.

Let $f \in \ker(\phi)$. Then, every coefficient of f is in I. Then, every monomial of f is in $\hat{I}$, so f is in $\hat{I}$ since $\hat{I}$ is closed under addition.

We thus conclude the proof. $\square$

Recall that every nonzero commutative ring contains a maximal ideal M. Let M be a maximal ideal in R. Let $\hat{M}$ be the ideal generated by M in $R[t]$. By the lemma above, we have that $R[t]/\hat{M}$ is isomorphic to $(R/M)[t]$.

Then, prime ideals of $R[t]$ containing $\hat{M}$ are in bijection with prime ideals of $(R/M)[t]$.

Thus, it suffices to show that every polynomial ring over a field F contains infinitely many prime ideals.

Since $F[t]$ is a Euclidean domain using the degree of the polynomial, we can simply reuse Euclid's proof.

Suppose not. Let $f_1, \ldots, f_n$ be the set of generators of the prime ideals. Consider $g = f_1 \cdot \cdots \cdot f_n + 1$. Since f_i divides g for some $i \in [n]$, we have that $f_i \mid 1$, which is a contradiction, since prime elements can't be units.